A cubic source for the vorticity maximum, and a recorded gap in the periodic continuation criterion

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Abstract

For a smooth divergence-free solution of the unforced periodic Navier–Stokes equations, let $M(t) = \|\omega(t)\|_{L^\infty}$. At a spatial maximum of $|\omega|$, transport vanishes and viscosity only lowers M, so $M' \le \omega \cdot S\omega / M$. The 7 October version claimed $\omega \cdot S\omega \le C_{\mathrm{CZ}} M^3$ from Calderón–Zygmund theory. That step is *not* proved: it needs $\|\nabla u\|_{L^\infty} \le C\|\omega\|_{L^\infty}$, which Calderón–Zygmund theory does not give. This corrected version replaces it by the classical log-corrected bound, and separately records the uncorrected inequality as a named hypothesis. The comparison ODEs, the ceiling under a quartic sink κM^4, and the implication from that sink to a finite Beale–Kato–Majda integral are checked in Lean. The sink itself remains open. Numerics at the vorticity maximum are reported and show no sign of a sink.

1 Setting

Let u be a smooth divergence-free solution of the unforced periodic Navier–Stokes equations on $\mathbb{T}^3$, with viscosity $\nu > 0$, and let $\omega = \nabla \times u$, $S = \frac{1}{2}(\nabla u + \nabla u^\top)$. Write $M(t) = \|\omega(t)\|_{L^\infty}$. The vorticity equation is

$$\partial_t \omega + (u \cdot \nabla)\omega = (\omega \cdot \nabla)u + \nu \Delta \omega, \qquad \omega \cdot (\omega \cdot \nabla)u = \omega \cdot S\omega.$$

2 The pairing at the maximum, and the corrected source bound

Proved (classical). At a spatial maximum x of $|\omega|$, $\omega \cdot (u \cdot \nabla)\omega = \frac{1}{2} u \cdot \nabla |\omega|^2 = 0$, and $\omega \cdot \Delta \omega = \frac{1}{2}\Delta|\omega|^2 - |\nabla \omega|^2 \le 0$. Hence, in the sense of upper Dini derivatives,

$$M\, M' \le \omega(x) \cdot S(x)\omega(x) \le \|S\|_{L^\infty} M^2. \tag{1}$$

The inconsistency in the 7 October version. That version stated that Biot–Savart is a Calderón–Zygmund operator bounded from L^∞ to BMO and *not* on L^∞, and in the next line used $\omega \cdot S\omega \le C_{\mathrm{CZ}} M^3$. The second statement requires $\|S\|_{L^\infty} \le C_{\mathrm{CZ}} \|\omega\|_{L^\infty}$, which is exactly the L^∞ bound the first statement denies. The Riesz transforms are unbounded on L^∞, so the bound $M' \le C_{\mathrm{CZ}} M^2$ was not proved and is withdrawn as a proved statement.

Replacement (a): the log-corrected bound (classical, not formalized here). The Beale–Kato–Majda / Kato–Ponce estimate gives, for $s > 3/2$,

$$\|\nabla u\|_{L^\infty} \le C\, M\Big(1 + \log^+ \frac{\|\omega\|_{H^s}}{M}\Big) + C\|\omega\|_{L^2},$$

so

$$M' \le CM^2\Big(1 + \log^+ \frac{\|\omega\|_{H^s}}{M}\Big) + C\|\omega\|_{L^2} M. \tag{2}$$

The logarithm involves a higher norm, which is controlled only through the BKM integral itself; (2) is therefore an ingredient of the BKM argument, not a closed ODE for M.

Replacement (b): conditional, under a named hypothesis.

Definition 1 (`PeriodicStrainLinfBound`, false)**.** There is $C \ge 0$ with $\|S(t,x)\| \le C\|\omega(t)\|_{L^\infty}$ for all smooth maximal solutions, all t below the lifespan, and all x.

This hypothesis is *false* for every $\nu > 0$ (Remark 1); it is recorded only to show what it would have given. Under it, $\omega \cdot S\omega \le CM^3$ (Lean: `pairing_le_of_periodicStrainLinfBound`) and therefore $M' \le CM^2$.

Remark 1 (`PeriodicStrainLinfBound` is false)*.* For every $\nu > 0$ there is no $C \ge 0$ with $\|S(t,x)\| \le C\|\omega(t)\|_{L^\infty}$ along all smooth maximal solutions (Lean: `not_periodicStrainLinfBound`, file `StrainLinfRefutation.lean`).

Proof. Step 1 (static failure). Let $f_K(s) = \frac{4}{\pi}\sum_{i<N} \frac{\sin(2\pi(2i+1)s)}{2i+1}$, $K = 2N-1$, and let u_K be the smooth, divergence-free, $\mathbb{Z}^3$-periodic, z-independent field with $\omega_K = \operatorname{curl} u_K = f_K(x) f_K(y)\, e_z$. Then $|f_K| \le 6$, so $\|\omega_K\|_{L^\infty} \le 36$ for all K (Lean: `vorticity_sup_norm_uK_le`; numerically $\|\omega_K\|_\infty \le 1.403$ for odd $K \ge 7$). The Fourier identity gives

$$S_{11}(u_K)(0) = \frac{16}{\pi^2} \sum_{i,j<N} \frac{1}{(2i+1)^2 + (2j+1)^2} \;\ge\; \frac{2}{\pi^2}\log(N+1)$$

(Lean: `strain00_origin`, `bcSum_ge_log`), and the sum diverges (`bcSum_unbounded`); asymptotically $\|S(u_K)(0)\| \sim \frac{2}{\pi}\log K$. Hence no C satisfies $\|S(u)(x)\| \le C\|\omega(u)\|_{L^\infty}$ for all smooth periodic divergence-free u (`not_staticStrainLinfBound`).

Step 2 (from data to solutions). Suppose the solution-level bound held with constant C. Take any smooth periodic datum u_0. Local existence gives $T^*(u_0) > 0$ (`periodicLocalExistence_of_nonneg`), and uniqueness gives a maximal solution u (`periodicMaximalSolution_of_uniqueness`, `periodicEnergyDifferenceEstimate`). Choose $0 < T < T^*$; the solution on $[0,T]$ satisfies $u(0) = u_0$. The bound at $t = 0$ then reads $\|S(u_0)(x)\| \le C\|\omega(u_0)\|_{L^\infty}$, i.e. the static bound, contradicting Step 1 (Lean: `not_periodicStrainLinfBound_of_static`). □

All inputs are theorems in the repository; no hypothesis Prop is used, and `#print axioms` reports only `propext`, `Classical.choice`, `Quot.sound`. Only the sharp Gibbs constant ≈ 1.39 is numerical, and it is not needed. The counterexample is classical (Bahouri–Chemin type); the explicit periodic version is published as B. S. Frohman, *Is* $\|S\|_{L^\infty} \le C\|\omega\|_{L^\infty}$*? A deep check of* `PeriodicStrainLinfBound`, Zenodo, doi:10.5281/zenodo.23904623 (Lean: `PeriodicNSContinuation/StrainLinf.lean`, commit `216ba1c`). The refutation concerns only the $t = 0$ uniform bound; the ν, t-dependent variant for $t > 0$ is a different statement and is not addressed here.

3 What the ODEs give (proved in Lean)

All statements in this section are abstract real-analysis lemmas in `CubicSource.lean`, with no `sorry` and only the standard axioms.

Proposition 1 (`quadratic_comparison`, `lifespan_lower_bound`)**.** *If $M > 0$ is differentiable and $M' \le CM^2$, then $1/M(t) \ge 1/M_0 - Ct$, so $M(t) \le M_0/(1 - CM_0 t)$ for $CM_0 t < 1$.*

This is a lifespan lower bound $T^* \ge 1/(CM_0)$ only. The comparison solution $y_0/(1 - Cy_0 t)$ blows up at $1/(Cy_0)$, so this cannot close BKM.

Proposition 2 (`cubic_comparison`)**.** *If $C \ge 0$ and $M' \le CM^2(1 + \log M)$, then $1/M(t)^2 \ge 1/M_0^2 - 2Ct$.*

Again only a lifespan lower bound: $\int^\infty dM/(M^2 \log M) < \infty$, so even a self-referential log correction does not prevent finite-time blowup of the comparison.

Proposition 3 (`fence_ceiling`, `sink_ceiling`)**.** *If M is continuous with right derivative $M' \le M^2(C - \kappa M)$, $\kappa > 0$, then $M \le \max(M_0, C/\kappa)$.*

4 The open gap: the quartic sink

Definition 2 (`PeriodicQuarticVorticitySink`, open, never assumed)**.** For every smooth maximal solution there are C and $\kappa > 0$ such that, below the lifespan, M is continuous with right derivative $M' \le M^2(C - \kappa M)$; equivalently, at the maximum, $\omega \cdot S\omega + \nu\, \omega \cdot \Delta\omega \le CM^3 - \kappa M^4$.

Theorem 1 (`bkm_bound_of_quarticSink`, proved)**.** *The quartic sink gives c with $M \le c$ and $\int_0^T M\, dt \le cT$ for every T below the lifespan. In particular the BKM integral is finite on $[0, T^*)$ whenever $T^* < \infty$, which by Beale–Kato–Majda would contradict finite lifespan.*

The sink is not derived from the unmodified vorticity equation. It is at least as strong as a uniform bound on the vorticity maximum, i.e. it is the problem restated as an inequality.

5 Three false fixes (unchanged)

(i) A formal certificate that expands the assumed rate $M(CM - \kappa M^2)$ into $CM^2 - \kappa M^3$ is ring arithmetic on a hypothesis. (ii) The weight $1 + \kappa M^2$ turns the source into $CM^3 + C\kappa M^5$: a degree-five source, not a sink. (iii) The logarithmic repair needs control of $\|\omega\|_{H^s}$ and is an integral criterion, not a derivation of the sink.

6 Numerics at the vorticity maximum (not proofs)

Pseudo-spectral RK4 runs on $\mathbb{T}^3$ (solver of `paper/viscous_balance_tests.py`; script `paper/cubic_source/cubic_source_tests.py`), random datum peaked at $|k| = 3$, snapshots every 0.05. At $x^* = \arg\max|\omega|$ we record $\sigma = \omega \cdot S\omega$, σ/M^3, $\|S\|_{L^\infty}/M$, $\nu\, \omega \cdot \Delta\omega/M^2$, and the proxy $\kappa_{\text{eff}} = (C_{\text{run}} M^3 - \sigma)/M^4$ with $C_{\text{run}} = \max \sigma/M^3$.

N	ν	seed	$t_{\max}$	M	σ/M^3	slope	$\|S\|_\infty/M$	min visc$/M^2$	min κ_{eff}
64	0.0025	1	8.00	9.1–43.0	-0.068–0.172	-0.84	0.34–0.53	-0.0705	2.2e-03
64	0.005	1	8.00	5.6–29.1	-0.040–0.141	-0.23	0.37–0.51	-0.1395	1.8e-03
64	0.005	2	8.00	5.8–28.7	-0.049–0.149	-0.46	0.35–0.61	-0.1324	1.5e-03
64	0.01	1	8.00	2.7–20.2	0.017–0.210	-0.34	0.40–0.53	-0.3205	6.2e-03
96	0.0025	1	8.00	10.4–49.7	-0.033–0.129	-0.28	0.33–0.55	-0.0829	4.3e-04
96	0.005	1	8.00	5.8–30.9	-0.014–0.139	-0.13	0.36–0.54	-0.1529	1.2e-03

Reading. (1) $\|S\|_{L^\infty}/M$ stays in about $[0.33, 0.61]$ over the observed range. So a ratio bound holds on these particular flows; the uniform bound `PeriodicStrainLinfBound` is nonetheless false in general (Remark 1). (2) σ/M^3 is mostly positive, with a maximum of 0.12–0.21 per run. Fitting $\log(\sigma/M^3)$ against $\log M$ over the positive snapshots gives slopes between -0.84 and -0.13. So the source is roughly cubic, or somewhat weaker than cubic, at the maximum. (3) σ is negative at some snapshots (0–22% of them; vortex compression at the maximum). These snapshots are intermittent: the fraction in the top quartile of M (0–20%) is comparable to the overall fraction, with no systematic trend with M. A sink κM^4 would require $\sigma/M^3 \le C - \kappa M$ at every large M; the data show no such ordering. (4) κ_{eff} is nonnegative only by the choice of C_{run}. Its minimum over the top quartile shrinks toward 0 (down to 4×10^{-4} at the finest run), which is the opposite of a uniform $\kappa > 0$. (5) The viscous term at the maximum is negative, of order 10–35 νM^2: a quadratic, not quartic, sink. *Caveats:* grids 64^3 and 96^3, all six runs complete to $t = 8$; peak vorticity only up to about 50; one or two seeds; decaying, unforced flow; snapshots every 0.05; the arg max is taken on the grid. Nothing here bears on behaviour near a putative singularity.

7 Record

Proved: the pairing (1); the comparison lemmas; the sink ceiling; sink $\Rightarrow$ finite BKM integral. **Refuted (proved in Lean):** `PeriodicStrainLinfBound` (Remark 1), so the conditional $M' \le CM^2$ under it is vacuous as a route; the log-corrected source (2) is classical and not formalized here. **Open:** `PeriodicQuarticVorticitySink`. No theorem in this note proves global regularity for the periodic Navier–Stokes equations.